Area and differential equations (2) (1)-1

Q1. If A_n is the area bounded by $y=x$ and $y=x^n$, $n \in N$, $A_2 \times A_3 \times A_4 \times \cdots \times A_n =$

(A) $\frac{1}{n(n+1)}$ (B) $\frac{1}{2^n(n)(n+1)}$ (C) $\frac{1}{2^{n-1}n(n+1)}$ (D) $\frac{1}{2^{n-2}n(n+1)}$

Q2. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it be $A(\alpha, f(\alpha))$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$, $(\alpha>0)$ the minimum area bounded by the line segments OA, OB & $f(x)$, where O is the origin is

(A) $\frac{\pi-1}{2}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{2}-1$ (D) $\frac{\pi+1}{2}$.

Q3. Minimum area of the region bounded by the curves $C_1: y=|x-4|$ & $C_2: y=-|x-a|+b$. Provided C_2 always passes through $(3,2)$ as a, b varies is

(A) $\frac{3}{2}$ (B) 0 (C) $\frac{3}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{2}}$.

Q4. If $|z-(4+4i)| = 4$, then the area of the region bounded by the foci of $z, iz, -z$ and iz in complex plane is

(A) $4(4-\pi)$ (B) $16(4-\pi)$ (C) $16(\pi-1)$ (D) $4(\pi-1)$

Q5. The area bounded by the curves $y = \sin^{-1}|\sin x|$ and $y = (\sin^{-1}|\sin x|)^2$, where $0 \le x \le 2\pi$ is

A) $\frac{1}{3} + \frac{\pi^2}{4}$ sq. units (B) $\frac{1}{6} + \frac{\pi^3}{8}$ sq. units

c) 2 sq. units (D) none of these

Q6. Solution of $y^2 dx + (x^2 - xy + y^2)dy = 0$ is (c is arbitrary constant)

(A) $\tan^{-1}\left(\frac{y}{x}\right) - \log y + c = 0$ (B) $2\tan^{-1}\left(\frac{y}{x}\right) - \log x + c = 0$

(C) $\log\left(y + \sqrt{x^2+y^2}\right) + \log y + c = 0$ (D) $\log\left(y - \sqrt{x^2+y^2}\right) - \log y + c = 0$

Q7. Through a point (x, y) of a curve passing the origin. lines L_1 & L_2 are drawn to the co-ordinate axes. The curve divides the rectangle formed by the lines L_1 & L_2 and the axes into two regions, one of which area is twice of the other. Then the curve may represent a family of

(A) circles (B) parabolas (C) Ellipses (D) Hyperbolas

Q8. Let $f : [0,2] \to R$ be a function which is continuous on $[0,2]$ and is differentiable on $(0,2)$ with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t})\,dt$ for $x \in [0,2]$. If $F'(x) = f'(x)$, for all $x \in (0,2)$ then $F(2)$ equals

(A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

Q9. Let f be a non negative continuous function defined on the interval $[0, \frac{\pi}{2}]$. If $\int_0^x ($ $f'(t) - \sin 2t)\,dt = \int_x^0 f(t) \tan t\,dt$ $\forall x \in (0, \frac{\pi}{2})$ and $f(0) = 1$, then

(A) $f(0) = 1$ is the absolute minimum value of f

(B) $f(0) = 1$ is the absolute maximum value of f

(C) $f(\frac{\pi}{4}) = \frac{3\sqrt{2}}{2} - 1$ (D) $f(\frac{\pi}{3}) = 1$

Q10. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ & $y = e - 1$

(A) $e - 1$ (B) $\int_1^e \ln(e + 1 - y)\,dy$

(C) $e - \int_0^1 e^x\,dx$ (D) $\int_1^e \ln y\,dy$.

Q11. Area of the region enclosed by the curves $C_1 : y = \operatorname{sgn} x$, $C_2 : x = 0$, $C_3 : y = x$ is not equal to

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Q12. Let $h = h(t)$ be a solution to the differential equation $\frac{dh}{dt} + 2ht + t^2$, then 16 $\lim_{t \to \infty} \left(\frac{h}{t}\right)$ is equal to

Q13. A spherical ball loses volume by evaporation at a rate proportional to its instantaneous area. If the diameter of the ball decreases from 4cm to 1cm in 3 days, how many more day(s) will take for the ball to be gone?

Q14. The rate at which a metal cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 290K and the substance cools from 370K 330 K in 10 min, then time required further to reach 295 K is 10α min, where $\alpha =$

Q15. The ratio in which $\frac{x}{4} + \frac{y}{3} = 1$ divides the region enclosed by $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is $K : 1 (K > 1)$. Then $[K - 2] =$ ...
(where $[\cdot]$ denotes GIF & take $\pi = 3.14$)

Q16 Area of the region enclosed by all the curves $x^2 + y^2 = 4$, $y = x^2 + x + 1$, $y = \left[\sin^2 \frac{x}{4} + \cos \frac{x}{4}\right]$ and x-axis (where $[\cdot]$ denotes GIF) is $\frac{K\pi}{6} + \frac{A\sqrt{3}}{2} - \frac{1}{B}$ (where $K, A, B \in N$). Then the value of $\left[K + A + \frac{B}{2}\right]$ equals to

Q17. The solution of the differential equation $\frac{dy}{dx} - ky = 0$, $y(0) = 1$ approaches to zero when $x \to \infty$, if

① $k = 0$ ② $k > 0$ ③ $k < 0$ ④ NOT.

Q18. The area of smaller part of the circle $x^2 + y^2 = a^2$ cut off by the line $x = \frac{a}{\sqrt{2}}$ in sq. units is

1) $\frac{a^2}{4}\left(\frac{\pi}{2} - 1\right)$ 2) $\frac{a^2}{2}\left(\frac{\pi}{2} - 1\right)$ 3) $a^2\left(\frac{\pi}{2} - 1\right)$ ④ $\frac{a^2}{4}(\pi - 1)$

Q19. The area of the region enclosed by the curves $y = x \log x$ & $y = 2x - 2x^2$ in sq. units is

1) $\frac{1}{12}$ 2) $\frac{5}{12}$ 3) $\frac{1}{4}$ 4) $\frac{7}{12}$.

Q20. The order of the differential equation whose general solution is given by $y = c_1 \cos(2x + c_2) - (c_3 + c_4)\, e^{x + c_5} + c_6 \sin(x - c_7)$ is

1) 3 2) 4 3) 5 4) 2

Q21. The area bounded by the curve $y = x^2(x+1)$ and the x-axis is

1) $\frac{2}{3}$ 2) $\frac{1}{4}$ 3) $\frac{1}{2}$ ④ $\frac{1}{12}$.

~~Q22. The~~

Q22. Area bounded by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ and the line $y = 1$ is

1) π sq. units 2) 2π sq. units 3) $\frac{\pi}{2}$ sq. units 4) NOT.

Q23. The area bounded by the curve $f(x) = x + \sin x$ and its inverse function between the ordinates $x = 0$ and $x = 3\pi$ is

1) 12π sq. units 2) 8π sq. units
3) 12 sq. units 4) 8 sq. units

Q24. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ in sq. units is

1) $4(\sqrt{2} - 1)$ 2) $4 - 2\sqrt{2}$ 3) $2(\sqrt{2} + 1)$ 4) $4 + 2\sqrt{2}$.

Q25. Let the straight line $x=b$ divide the area enclosed by $y=(1-x)^2$, $y=0$ and $x=0$ into two parts R_1 $(0 \le x \le b)$ & R_2 $(b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. then $4b =$

1) 3 2) 2 3) 8 4) 1

Q26. The area of the region enclosed by the curves $y=x^2$ and $y=\sqrt{|x|}$ in sq. units is

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{1}{6}$ 4) 1

Q27. The ratio in which $y=x^2$ divides the region bounded by the curve $y=\sin\frac{\pi}{2}x$ and the x-axis as x varies from 0 to 1 is

1) $2:\pi$ 2) $1:3$ 3) $3:\pi$ 4) $(6-\pi):\pi$

passage (28-29)

Let $y=f(x)$ satisfy the differential equation

$$f(x) = (e^x + e^{-x})\cos x - 2x - \int_0^x (x-t) f'(t)\,dt$$

Q28. If y satisfies the differential equation $\frac{dy}{dx} + y = e^x g(x) + e^{-x} g(-x)$ then $g(x)$ is ---

(A) $\sin x + \cos x$ (B) $\sin x - \cos x$ (C) $\cos x - \sin x$ (D) $-\sin x - \cos x$.

Q29: The value of $f'(\pi) + f''(\pi) =$ ---

(A) -1 (B) 0 (C) 1 (D) 2.

passage (30-31)

The boundaries of quadrilateral Q are given by $1 + x^2y^2 - x^2 - y^2 = 0$. R is a region belonging to Q such that every point 'P' of 'R' is nearer to the origin O(0,0) than to any edge of 'Q'. Now answer the following questions:

Q30.

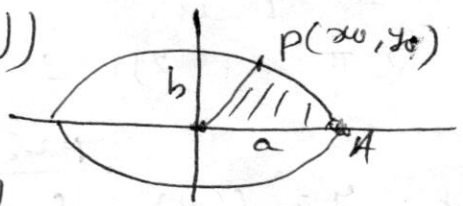
Q30. Area of the region R is ... sq. units

(A) $\frac{1+6\sqrt{2}}{3}$ (B) $\frac{3\sqrt{2}-1}{3}$ (C) $\frac{16\sqrt{2}-20}{3}$ (D) $\frac{20\sqrt{2}-16}{5}$

Q31. Maximum and least values of distance AB where A & B points on boundries of R and Q respectively are M and L respectively then $\sqrt{\frac{M}{L}}$ = ---

a) 2 b) $\frac{4}{\sqrt{3}}$ (c) $\frac{2}{\sqrt{3}}$ d) $\frac{\sqrt{5}}{3}$.

Q32. An ellipse shown here has equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a>b$. If $P(x_0, y_0)$ is any point on it and area of curvilinear part OPA is denoted by Δ_0. Then which of the following statements can be wrong?



(A) $P: \left(a\cos\left(\frac{2\Delta_0}{ab}\right), b\sin\left(\frac{2\Delta_0}{ab}\right)\right)$

(B) $P: \left(a\cos\left(\frac{2\sqrt{\Delta_0}}{ab}\right), b\sin\left(\frac{2\sqrt{\Delta_0}}{ab}\right)\right)$

(C) $P: \left(a\cos\left(\frac{ab}{2\Delta_0}\right), b\sin\left(\frac{ab}{2\Delta_0}\right)\right)$

(D) $P: \left(a\cos\left(\frac{\sqrt{ab}}{2\Delta_0}\right), b\sin\left(\frac{\sqrt{ab}}{2\Delta_0}\right)\right)$.

Q33. The gradient of curve passing through (4,0) is given by $\frac{dy}{dx}-\frac{y}{x}+\frac{5x}{(x+2)(x-3)}=0$. If the point (5, a) lies on the curve then

(A) Domain of $y(x)$ is $(-2,3)$ (B) Domain of $y(x)$ is $(-\infty,-2)\cup(3,\infty)$ (C) $a=5\ln\left(\frac{12}{7}\right)$

(D) $a=-5\ln\left(\frac{12}{7}\right)$.

Q34. If $y=f(x)$ satisfies the differential equation $(\sin x)\frac{dy}{dx}-y\cos x+\frac{\sin 2x}{x^2}=0$ & such that $y\to 0$ as $x\to\infty$ then which of the following statements is possible?

(A) $\lim_{x\to\infty} f(x)=1$ (B) least value of $f(x)$ in $\left[\frac{\pi}{4},\frac{\pi}{3}\right]$ is $\frac{3\sqrt{3}}{2\pi}$ (C) $\int_0^{\pi/2} f(x)\,dx>1$

(D) $f(x)$ is an even function.

Q 35. Let $f(x)$ and $g(x)$ be two continuous functions on $a \le x \le b$ intersecting at only one point in (a,b) & let $p(x) = \max\{f(x), g(x)\}$ and $q(x) = \min\{f(x), g(x)\}$ then the area bounded by the curves $y = p(x)$, $y = q(x)$ and ordinates $x = a$ & $x = b$ is given by

A) $\int_a^b (f(x) - g(x))\,dx$ B) $\int_a^b (p(x) - q(x))\,dx$

C) $\int_a^b |f(x) - g(x)|\,dx$ D) $\int_a^b |p(x) - q(x)|\,dx$

Q 36. Suppose $y = f(x)$ and $y = g(x)$ are two functions whose graphs intersect at three points $(0,4)$, $(2,2)$ & $(4,0)$ with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 (f(x) - g(x))\,dx = 10$ & $\int_2^4 (g(x) - f(x))\,dx = 5$. the area between the curves for $0 < x < 2$ is

A) 8 B) 10 C) 15 D) NOT.

Q 37. Five regions are defined by $C_1 : |x + y| \le 1$, $C_2 : |x - y| \le 1$, $C_3 : |x| \le \frac{1}{2}$, $C_4 : |y| \le \frac{1}{2}$, $C_5 : 3x^2 + 3y^2 = 1$: then

A) The area bounded by C_1, C_2 which doesn't contain area of C_5 is $2 - \frac{\pi}{3}$

B) The part of area of C_5 which doesn't contain area of C_3 and C_4 is $\frac{2\pi}{9} - \frac{1}{\sqrt{3}}$

C) The part of area bounded by C_1, C_2 but not bounded by C_3 & C_4 is

D) Area bounded by C_1, C_2, C_3, C_5 but not by C_4 is $\frac{\pi}{2} - \frac{1}{\sqrt{3}}$.

15 (8)

Q4. If $|z-(4+4i)|=4$, then the area of the region bounded by the foci of z, iz, $-z$ and iz in complex plane is

(A) $4(4-\pi)$ (B) $16(4-\pi)$ (C) $16(\pi-1)$ (D) $4(\pi-1)$.

Answers

(1) - D (2) A (3) - A, (4) - B, 5 - D $\left(\frac{\pi^3}{6}-\frac{\pi^2}{2}+\frac{4}{3}\right)$

6 - A, 7 - B, 8 - B, 9 - A, 10 - B, C, D, 11 - ~~A~~ BCD

12 - 8, 13 - 1, 14 - 4, 15 - 8, 16 - 9

17 - C, 18 - 2, 19 - 2, 20 - 3, 21 - 4

22 - 2, 23 - 3, 24 - 2, 25 - 2, 26 - 1

27 - 4, 28 - C, 29 - B, 30 - Ans $\left(\frac{2\sqrt{2}-20}{3}\right)$ $\frac{16\sqrt{2}-20}{3}$ (C)

31 - a, 32 - A, 33 - B, D, 34 - B, C, D

35 - B, C, D, 36 - C, 37 - A, C